

18.2 Adjusted treatment means

A covariance analysis is usually reported as a table of treatment means “adjusted” for the covariate. The raw mean $\bar{y}_k$ estimates the mean response of treatment k at whatever covariate values its units happened to have; the adjusted mean estimates it at a covariate value common to all treatments. This section defines adjusted means, gives their standard errors, and shows how to compare them.

18.2.1 Definition

Take the one-way covariance model with g groups of sizes $n_1, \dots, n_g$ and q covariates,

$$E(y_i) = \mu_k + \mathbf{z}_{(i)}^\top \boldsymbol{\gamma} \quad \text{for unit } i \text{ in group } k, \quad (18.2.1)$$

where $\mathbf{z}_{(i)} \in \mathbb{R}^q$ is the covariate vector of unit i , and write $\bar{\mathbf{z}}_k$ for the mean covariate vector of group k and $\bar{\mathbf{z}}$ for the overall mean. We assume Eq. 18.1.3, and write $\mathbf{E}_{zz}$ and $\hat{\boldsymbol{\gamma}}$ as in Theorem 18.1.1.

Definition 18.2.1. Adjusted treatment means

For a fixed covariate vector $\mathbf{z}_0 \in \mathbb{R}^q$, the **adjusted mean** of treatment k at $\mathbf{z}_0$ is $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$, the mean response of a unit with covariate $\mathbf{z}_0$ under treatment k . Its estimate is

$$\hat{\mu}_k(\mathbf{z}_0) = \bar{y}_k - (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \hat{\boldsymbol{\gamma}}. \quad (18.2.2)$$

When no value is named, $\mathbf{z}_0 = \bar{\mathbf{z}}$, the overall mean of the covariates, and $\hat{\mu}_k(\bar{\mathbf{z}})$ is called simply the adjusted mean of treatment k .

The fitted line (or plane) of treatment k passes through $(\bar{\mathbf{z}}_k, \bar{y}_k)$ with slope $\hat{\boldsymbol{\gamma}}$; the adjusted mean is its height at $\mathbf{z}_0$. In Figure 18.1.1 the filled symbols on the dashed line are the adjusted means.

The adjusted mean is estimable: $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$ is the sum of $\mu_k + \bar{\mathbf{z}}_k^\top \boldsymbol{\gamma}$, the mean of the fitted values of group k , and $(\mathbf{z}_0 - \bar{\mathbf{z}}_k)^\top \boldsymbol{\gamma}$, a multiple of the estimable $\boldsymbol{\gamma}$ (Theorem 18.1.1(b)). By Theorem 18.1.1(c) with $\boldsymbol{\rho}$ the scaled indicator of group k , the least squares estimate of μ_k is $\bar{y}_k - \bar{\mathbf{z}}_k^\top \hat{\boldsymbol{\gamma}}$, which gives Eq. 18.2.2.

18.2.2 Standard errors

Proposition 18.2.1. Moments of adjusted means

In the model Eq. 18.2.1 with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and for a fixed $\mathbf{z}_0$:

- (a) $\bar{y}_1, \dots, \bar{y}_g$ are uncorrelated with $\hat{\boldsymbol{\gamma}}$, and

$$\text{Var}(\hat{\mu}_k(\mathbf{z}_0)) = \sigma^2 \left(\frac{1}{n_k} + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \mathbf{z}_0) \right); \quad (18.2.3)$$

- (b) for $k \neq l$, $\text{Cov}(\hat{\mu}_k(\mathbf{z}_0), \hat{\mu}_l(\mathbf{z}_0)) = \sigma^2 (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_l - \mathbf{z}_0)$;

- (c) for a contrast $\psi = \sum_k c_k \mu_k$, the estimate $\hat{\psi} = \sum_k c_k \hat{\mu}_k(\mathbf{z}_0) = \sum_k c_k \bar{y}_k - \bar{\mathbf{z}}_c^\top \hat{\boldsymbol{\gamma}}$, with

$\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k$, does not depend on $\mathbf{z}_0$, and

$$\text{Var}(\hat{\psi}) = \sigma^2 \left(\sum_k \frac{c_k^2}{n_k} + \bar{\mathbf{z}}_c^\top \mathbf{E}_{zz}^{-1} \bar{\mathbf{z}}_c \right); \quad (18.2.4)$$

- (d) with $\mathbf{z}_0 = \bar{\mathbf{z}}$, the weighted average of the adjusted means is the grand mean: $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_i y_i$.

Proof. Let $\mathbf{M}$ project onto the span of the group indicators. The group mean $\bar{y}_k$ is $\mathbf{u}_k^\top \mathbf{Y}$ with $\mathbf{u}_k$ the indicator of group k divided by n_k , a vector in $\mathcal{C}(\mathbf{M})$. The estimate $\hat{\mathbf{y}} = \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ is a linear function of $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$. Hence $\text{Cov}(\bar{y}_k, \hat{\mathbf{y}}) = \sigma^2 \mathbf{u}_k^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z} \mathbf{E}_{zz}^{-1} = \mathbf{0}$, since $(\mathbf{I} - \mathbf{M}) \mathbf{u}_k = \mathbf{0}$. Then (a) and (b) follow from $\text{Var}(\bar{y}_k) = \sigma^2/n_k$, $\text{Cov}(\bar{y}_k, \bar{y}_l) = 0$ and $\text{Cov}(\hat{\mathbf{y}}) = \sigma^2 \mathbf{E}_{zz}^{-1}$ (Theorem 18.1.1(e)). For (c), $\sum_k c_k = 0$ removes $\mathbf{z}_0$, and the variance follows in the same way. For (d), $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_k n_k \bar{y}_k - (\sum_k n_k \bar{\mathbf{z}}_k - n \bar{\mathbf{z}})^\top \hat{\mathbf{y}}$, and the bracket is zero. ■

Adjusted means are correlated, because their errors share $\hat{\mathbf{y}}$. Their variances grow with the distance of $\bar{\mathbf{z}}_k$ from $\mathbf{z}_0$: an adjusted mean is a prediction from the treatment's line, and for a group whose covariate values barely overlap the others' it rests on extrapolation, as in [Section 12.3](#). Differences do not depend on $\mathbf{z}_0$; only the common level and the individual standard errors do. The overall mean $\bar{\mathbf{z}}$ keeps the adjusted means on the scale of the data (d) and minimizes $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$ ([Exercise \(B1\)](#)), but any stated value, such as a clinically meaningful baseline, is legitimate.

18.2.3 Comparing adjusted means

Pairwise differences are contrasts, so [Eq. 18.2.4](#) gives

$$\text{se}(\hat{\mu}_k - \hat{\mu}_l) = s \sqrt{\frac{1}{n_k} + \frac{1}{n_l} + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)}, \quad s^2 = \frac{\text{SSE}}{n - g - q}, \quad (18.2.5)$$

and a single comparison uses the t interval of Theorem 12.1.1 with $n - g - q$ degrees of freedom. For simultaneous comparisons the correlations matter.

- (i) *Scheffé*. The adjusted treatment F test of Theorem 18.1.1(f) is a test of $\mu_1 = \dots = \mu_g$, a $(g - 1)$ -dimensional space of estimable contrasts. By Theorem 13.2.2, the intervals $\hat{\psi} \pm \sqrt{(g - 1)F_\alpha(g - 1, n - g - q)} \text{se}(\hat{\psi})$ have simultaneous coverage exactly $1 - \alpha$ over *all* contrasts ψ . Some interval excludes zero iff the adjusted F test rejects (Corollary 13.2.3).
- (ii) *Bonferroni*. For a planned list of m comparisons, t intervals at level α/m have coverage at least $1 - \alpha$ whatever the correlations (Theorem 13.4.1).
- (iii) *Tukey-type intervals*. The half-width $(q_\alpha(g, n - g - q)/\sqrt{2}) \text{se}(\hat{\mu}_k - \hat{\mu}_l)$ copies the Tukey–Kramer rule [Eq. 13.3.3](#). It is exact when the covariate means are equal and the groups balanced ([Exercise \(C1\)](#)). Otherwise its coverage is only approximately $1 - \alpha$: Hayter's theorem covers unequal group sizes in the one-way layout, but not the correlations that the covariate introduces. When the covariate means are close, as they usually are after randomization, the departure is small ([Exercise \(C2\)](#) asks you to measure it). When they are far apart, use Scheffé's intervals, or compute the exact multiplier for the given

correlation matrix by simulation, as [Section 13.3](#) did for Tukey–Kramer.

Example. Adjusted means for the four formats

In Example (Four study formats with a pretest), the overall pretest mean is $\bar{x} = 59.92$. Format 2 had pretest mean 57.2, so its adjusted mean is

$$75.15 - 0.8670(57.2 - 59.92) = 75.15 + 2.36 = 77.51.$$

The four adjusted means are 70.97, 77.51, 69.11 and 77.59, averaging to the grand mean 73.80, with standard errors 2.02, 2.03, 2.01 and 2.01. These barely differ, because the pretest means are close to $\bar{x}$ compared with the within-group standard deviation of the pretest, 10.06.

The six pairwise comparisons, with $s = 6.34$ on 35 degrees of freedom, are as follows. The Tukey-type multiplier is $q_{0.05}(4, 35)/\sqrt{2} = 2.697$, and Scheffé's is $\sqrt{3F_{0.05}(3, 35)} = 2.936$.

formats	raw difference	adjusted difference	se	Tukey-type half-width	Scheffé half-width
1 – 2	–2.29	–6.54	2.88	7.77	8.46
1 – 3	3.77	1.86	2.85	7.67	8.36
1 – 4	–5.23	–6.62	2.84	7.66	8.34
2 – 3	6.06	8.40	2.85	7.69	8.37
2 – 4	–2.94	–0.08	2.86	7.71	8.39
3 – 4	–9.00	–8.48	2.84	7.65	8.33

Both methods declare formats 2 and 4 better than format 3 and resolve nothing else, although the 2 – 3 comparison clears Scheffé's bound by only 0.03. Without the covariate the standard error of a difference would be $10.73\sqrt{2/10} = 4.80$. The raw difference between formats 3 and 4, –9.00, would then have had a Tukey half-width near 13, and nothing would have been resolved.

Adjustment moved comparisons in both directions: format 2's advantage over format 1 grew from 2.29 to 6.54, while format 4's lead over format 3 shrank from 9.00 to 8.48.

```
import itertools
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.ROUND(rng.normal(60, 10, n))             # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.ROUND(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)
Z = np.eye(g)[group]                         # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k    # group means
```

```

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                                # slope from within-group
            variation only
sse = E_yy - E_xy ** 2 / E_xx                      # ANCOVA residual sum of
            squares
nu = n - g - 1

s2 = sse / nu
x0 = x.mean()                                     # the covariate value for
            adjustment
adj = y_bar - beta * (x_bar - x0)                  # adjusted means
se_adj = np.sqrt(s2 * (1 / n_k + (x_bar - x0) ** 2 / E_xx))
pairs = list(itertools.combinations(range(g), 2))
q = stats.studentized_range.ppf(0.95, g, nu) / np.sqrt(2)  # Tukey-Kramer
            multiplier
c_s = np.sqrt((g - 1) * stats.f.ppf(0.95, g - 1, nu))      # Scheffe multiplier
for k, l in pairs:
    d = adj[k] - adj[l]
    se_d = np.sqrt(s2 * (1 / n_k[k] + 1 / n_k[l] + (x_bar[k] - x_bar[l]) ** 2 / E_xx))
    print(f"[k + 1]-[l + 1]: raw {y_bar[k] - y_bar[l]:6.2f} adjusted {d:6.2f} se {
        se_d:.2f}")
        f" Tukey-type +-{q * se_d:.2f} Scheffe +-{c_s * se_d:.2f}")
print("adjusted means", adj.round(2), " standard errors", se_adj.round(2))

```

[Run this code online](#)

18.2.4 Adjusted means in other designs

For any design part, the adjusted version at $\mathbf{z}_0$ of an estimable $\lambda^\top \beta$, such as a treatment mean averaged over blocks, is $\lambda^\top \beta + \mathbf{z}_0^\top \gamma$, estimated by $\rho^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\gamma}) + \mathbf{z}_0^\top \hat{\gamma}$ with variance from Theorem 18.1.1(e). Software calls these **least squares means**. In unbalanced multi-way data the choice of λ , equal or count-proportional weights over cells, matters as much as $\mathbf{z}_0$ (Chapter 17, Theorem 17.2.2); it is safest to write the intended function of the cell means (Definition 17.1.1) explicitly.

Remark (The adjusted mean is not a prediction interval). The standard error Eq. 18.2.3 describes the uncertainty about the *mean* response of treatment k at $\mathbf{z}_0$. A new unit with covariate $\mathbf{z}_0$ given treatment k will vary around that mean with variance σ^2 as well. Its prediction interval uses $s\sqrt{1 + 1/n_k + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \mathbf{z}_0)}$, as in Theorem 12.3.1.

18.2.5 Exercises

18.2.5.1 A. Check your understanding

Exercise. A1

From the numbers in Example (Four study formats with a pretest), compute the adjusted means of formats 1 and 3 by hand and check them against the values in [Example \(Adjusted means for the four formats\)](#).

Exercise. A2

Software that fits the model $E(y) = \mu_k + \beta x$ reports $\hat{\mu}_k$, the adjusted mean at $x_0 = 0$. For a pretest scored from 0 to 100, what does this number describe? Show that its standard error can be far larger than that of the adjusted mean at $\bar{x}$, although both give the same differences.

18.2.5.2 B. Practice**Exercise. B1**

Show that $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$ is minimized over $\mathbf{z}_0$ by $\mathbf{z}_0 = \bar{\mathbf{z}}$.

Solution. By [Eq. 18.2.3](#), the sum is $\sigma^2(g + \sum_k n_k (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \mathbf{z}_0))$. Writing $\bar{\mathbf{z}}_k - \mathbf{z}_0 = (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + (\bar{\mathbf{z}} - \mathbf{z}_0)$ and using $\sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) = \mathbf{0}$, the cross terms vanish and the sum becomes

$$\sigma^2 \left(g + \sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}})^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + n (\bar{\mathbf{z}} - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}} - \mathbf{z}_0) \right).$$

The last term is nonnegative, and zero iff $\mathbf{z}_0 = \bar{\mathbf{z}}$, because $\mathbf{E}_{zz}$ is positive definite.

Exercise. B2

A common alternative to covariance adjustment analyses the *change* $y - x$. Show that this is the covariance model with β fixed at 1. Using Proposition 18.1.2's setting with τ^2 the variance of x , find the variance of the difference of mean changes between two groups, and show that it exceeds the variance of the adjusted difference (ignoring the $1/(N - g - 2)$ term) unless $\beta = 1$.

Solution. The mean change in group k is $\bar{y}_k - \bar{x}_k = \mu_k + (\beta - 1)\bar{x}_k + \bar{\varepsilon}_k$. The difference of two groups has variance $(2/m)\{(\beta - 1)^2\tau^2 + \sigma^2\}$. The adjusted difference, with the slope known, has variance $2\sigma^2/m$. The change-score analysis is therefore worse by $(2/m)(\beta - 1)^2\tau^2$, which is zero only when $\beta = 1$. It beats the unadjusted analysis, which has $(\beta - 1)^2$ replaced by β^2 , only if $\beta > 1/2$. Covariance adjustment estimates the slope rather than assuming one.

18.2.5.3 C. Going deeper**Exercise. C1**

Suppose all groups have size m and the covariate means $\bar{\mathbf{z}}_1, \dots, \bar{\mathbf{z}}_g$ are equal. Show that the adjusted differences $\hat{\mu}_k - \hat{\mu}_l$ are the raw differences $\bar{y}_k - \bar{y}_l$, that these are independent of s^2

from the covariance model, and that the Tukey intervals built from s and $q_\alpha(g, n - g - q)$ have simultaneous coverage exactly $1 - \alpha$.

Solution. With equal covariate means, $\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k = \mathbf{0}$ for every contrast, so the adjusted contrast equals the raw one and has variance $\sigma^2 \sum_k c_k^2 / m$ by Eq. 18.2.4. The group means are functions of $\mathbf{MY}$. The residual vector $(\mathbf{I} - \mathbf{M}_W)\mathbf{Y}$ is orthogonal to $\mathcal{C}(\mathbf{M}_W) \supseteq \mathcal{C}(\mathbf{M})$, so under normality s^2 is independent of $\bar{y}_1, \dots, \bar{y}_g$ (Theorem 4.4.4), and $(n - g - q)s^2/\sigma^2 \sim \chi^2(n - g - q)$. The means are independent normal with common variance σ^2/m . The maximum of $|(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)|/(s/\sqrt{m})$ therefore has the studentized range distribution with parameters g and $n - g - q$, and the proof of Theorem 13.3.2 applies unchanged.

Exercise. C2

For four groups of ten and one covariate, simulate the simultaneous coverage of the Tukey-type intervals of Example (Adjusted means for the four formats) when the covariate means are held at (62, 57, 60, 61) and when they are held at (70, 50, 60, 60), with the within-group covariate variation as in the example. How far from 0.95 is the coverage in each case?